Chapter 3
Vector Spaces

3.1 Vector and Vector Space Axioms

Logic of Chapter 3

$A\boldsymbol{x} = \boldsymbol{b}$

$x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2 + \cdots + x_n\,\boldsymbol{a}_n = \boldsymbol{b}$

Consistency

Consistency: $\boldsymbol{b}$ in the **column space of** $\boldsymbol{A}$: $col\ \boldsymbol{A}$

Row echelon form

What the solution set looks like: forms a **subspace.**

Invertibility of A

How large is the solution set
basis, dimension, rank,
The Rank–Nullity Theorem,
$\text{rank}(A) + \dim N(A) = n$

$Det(A)$

Are there solutions?
How to find all solutions?

Changing basis

Objective

- n-dimensional real vector space
  - 二维、三维空间是最经典的向量空间：向量可以“加减”和“缩放”
- Vector Space Axioms
  - 向量空间的抽象定义，只需定义“向量加法”和“标量乘法”，并且满足一系列公理，就是向量空间
- Other examples of vector spaces
  - “矩阵”、“函数”、“多项式”等也可以形成向量空间，因为加减法、标量乘法满足公理

$\mathbb{R}^n$

Definition

The set of all column n-vectors $[u_1, u_2, \ldots, u_n]^T$ is called ***n-dimensional real space*** and is denoted $\mathbb{R}^n$.

u_i is the ***i*-th component** of $[u_1, u_2, \ldots, u_n]^{\mathrm{T}}$.

Example

$\mathbb{R}^4$ is the sets of all 4x1 column vectors. For example, $[1, 2, 3, 4]^T$ and $[-1, 3, 5.2, 0]^T$ are in $\mathbb{R}^4$.

$\mathbb{R}^5$ is the set of all 5x1 column vectors. For example, $[-1, 2, 0, 3, 9]^T$ is in this set.

Addition and Scalar Multiplication

Definition

Let $\mathbf{u} = [u_1, u_2, \ldots, u_n]^T$ and $\mathbf{v} = [v_1, v_2, \ldots, v_n]^T$ be two elements of $\mathbb{R}^n$. We say that **u** and **v** are **equal** if $u_1 = v_1$, …, $u_n = v_n$.

Let $\mathbf{u} = [u_1, u_2, \ldots, u_n]^T$ and $\mathbf{v} = [v_1, v_2, \ldots, v_n]^T$ be two elements of $\mathbb{R}^n$ and let *c* be a scalar. Addition and scalar multiplication are defined as

加法

Addition: $\mathbf{u} + \mathbf{v} = [u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n]^T$

标量乘法

Scalar multiplication : $c\mathbf{u} = [cu_1, cu_2, \ldots, cu_n]^T$

Note.

在加法下闭合

(1) **u**, **v**$\in \mathbb{R}^n \Rightarrow$ **u**+**v** $\in \mathbb{R}^n$ ($\mathbb{R}^n$ is **closed** under addition)

(2) **u**$\in \mathbb{R}^n$, $c \in$ **R** $\Rightarrow c$**u** $\in \mathbb{R}^n$ ($\mathbb{R}^n$ is **closed** under scalar multiplication)

在标量乘法下闭合

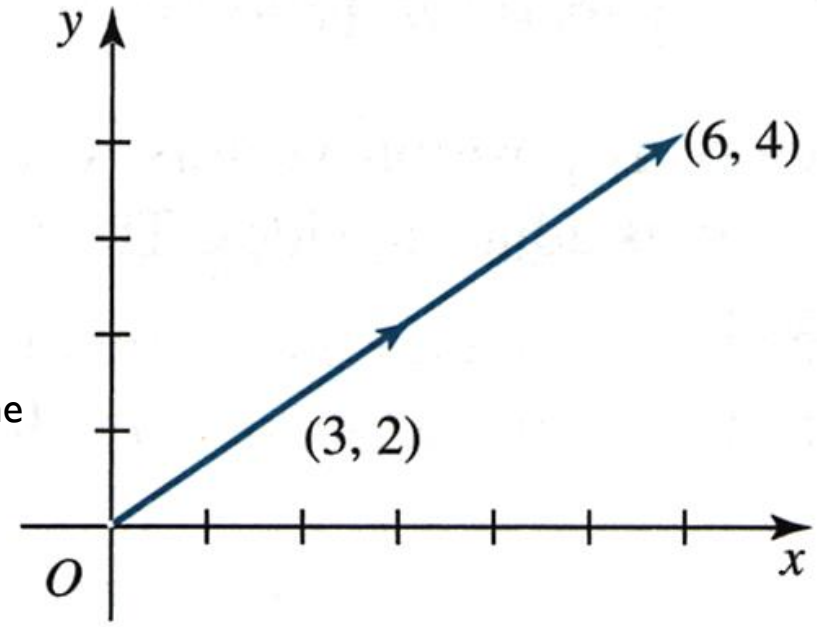
Scalar multiplication example

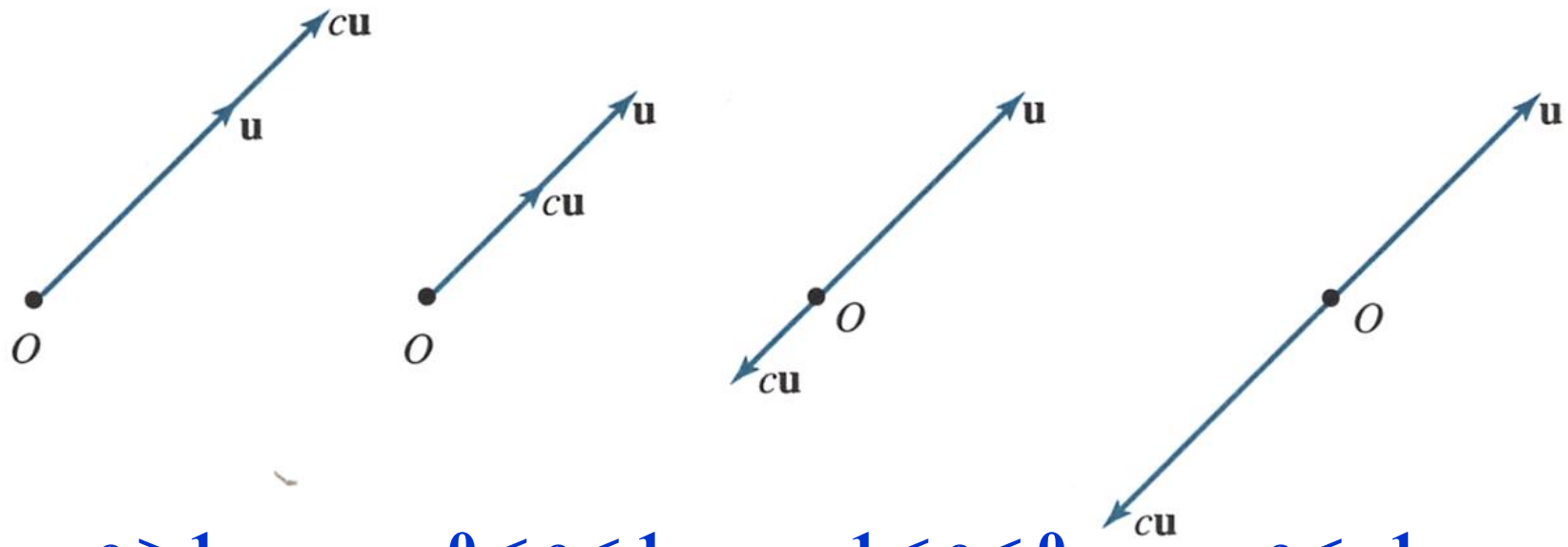
Consider the scalar multiple of the vector $[3,2]^T$ by 2, we get

$$2[3,2]^T = [6,4]^T$$

Observe that $[6,4]^T$ is a vector in the same direction as $[3,2]^T$, but 2 times further away.

(Distracting question on the side: what is the "length" of the vector $[3,2]^T$? How should you calculate the "length" of an n-vector? More about that in the future.)





c > 1 **0 < c < 1** **–1 < c < 0** **c < –1**

- The direction of *c***u** will be
  - the same as the direction of **u** if $c > 0$,
  - the opposite direction to **u** if $c < 0$.
- The "length" of c**u** is |c| times the length of **u**.
- Property: multiplying **u** by β times, and then by α times, is the same as multiplying **u** by $\alpha \times \beta$ times.

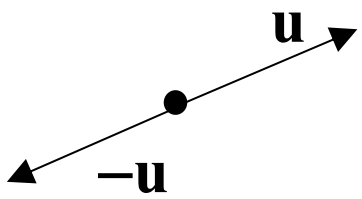
Zero vector, negative vector

The vector $[0,0,\ldots,0]^T$, having *n* zero components, is called the **zero vector** of $\mathbb{R}^n$ and is denoted **0**.
Property: for any vector v, **0 + v = v, v + 0 = v.**

Negative Vector

The vector (–1)**u** is written –**u** and is called the **negative of u**. It is a vector having the same magnitude as **u**, but lies in the opposite direction to **u**.

Property: for any vector **u**,
u + (–**u**) = **0** = (- **u**) + **u**

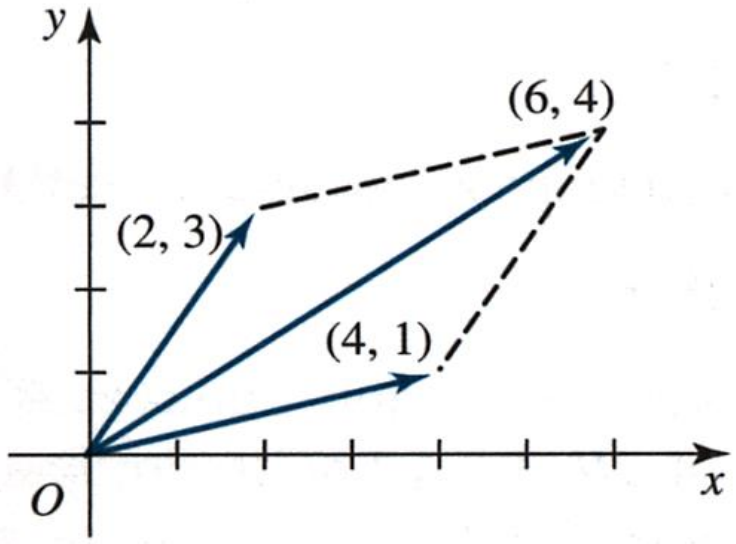


Addition and subtraction

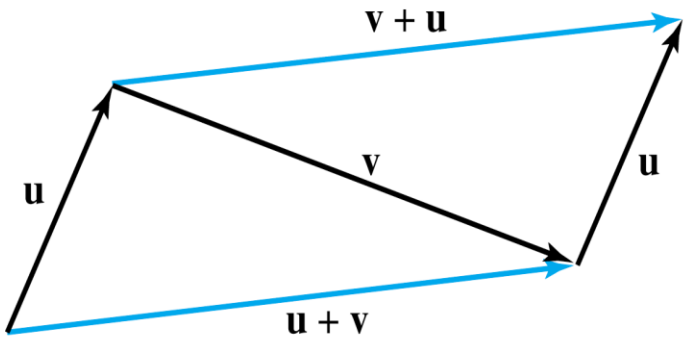
- if **u** and **v** are 2 vectors, then **u+v** is another vector:
  - Algebraically, **u+v** = $\begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}$
  - Geometrically, **u** + **v** is the diagonal of the parallelogram（平行四边形） defined by **u** and **v**.
- $\mathbf{u} - \mathbf{v}$ is the vector pointing from $\mathbf{v}$ to $\mathbf{u}$.

Example: Consider the vector (4, 1 and (2, 3), then

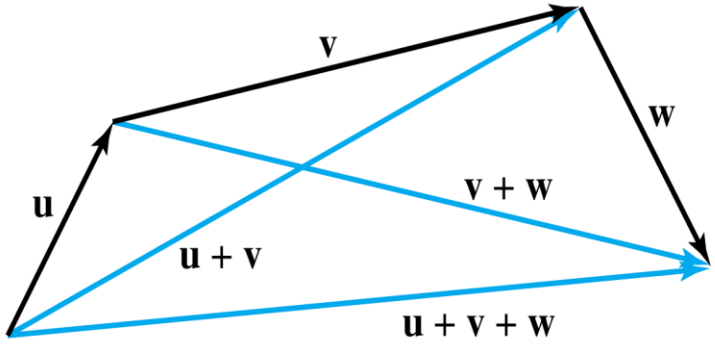
$$\begin{bmatrix} 4 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}.$$



Properties of vector additions



$\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u}.$



$(\mathbf{u}+\mathbf{v})+\mathbf{w}=\mathbf{u}+(\mathbf{v}+\mathbf{w}).$

Vector addition and scalar multiplications have many good properties:

A1. **x** + **y** = **y** + **x** for $\forall\, x, y \in V$

A2. (**x** + **y**) + **z** = **x** + (**y** + **z**), for all **x, y, z** in V

A3. There exist an element **0** $\in V$, such that **x** + **0** = **0** + **x** = **x** holds for any **x**

A4. for any x, there exist an (negative) element, -**x**, such that x + (–**x**) = **0**

A5. α(**x** + **y**) = α**x** + α**y**

A6. $(\alpha + \beta)$ **x** = α**x** + β **x**

A7. $(\alpha\beta)$**x** = $\alpha(\beta$**x** $)$

A8. 1 **x** = **x**

Question: where else have you seen similar properties of addition and scalar multiplication?

Vector Space – Abstract definition

Definition

Let V be a set on which the operations of *addition* and *scalar multiplication* are defined. By this we mean that,

C1 with each pair of elements **x** and **y** in V, there exists a unique element **x** + **y** that is **also in** $\boldsymbol{V}$,

C2 with each element **x** in V and each scalar α, there exists a unique element $\alpha \circ \mathbf{x}$ that is **also in** $\boldsymbol{V}$.

Then V together with operations addition and scalar multiplication is called a ***vector space*** if the following axioms are satisfied (A1-A8):

A1. $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$ for $\forall\, \boldsymbol{x}, \boldsymbol{y} \in V$

A2. $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$, for all **x, y, z** in V

A3. There exist an element $\mathbf{0} \in V$, such that $\mathbf{x} + \mathbf{0} = \mathbf{0} + \mathbf{x} = \mathbf{x}$ holds for any **x**

A4. for any x, there exist an (negative) element, -**x**, such that $\boldsymbol{x} + (-\mathbf{x}) = \mathbf{0}$

A5. $\alpha \circ (\mathbf{x} + \mathbf{y}) = \alpha \circ \mathbf{x} + \alpha \circ \mathbf{y}$

A6. $(\alpha + \beta) \circ \mathbf{x} = \alpha \circ \mathbf{x} + \beta \circ \mathbf{x}$

A7. $(\alpha\beta) \circ \mathbf{x} = \alpha \circ (\beta \circ \mathbf{x})$

A8. $1 \circ \mathbf{x} = \mathbf{x}$

Review: Matrix Operations

We have seen something similar to the 8 axioms of vector spaces when we defined matrix addition and scalar multiplication in Section 1.3.

Theorem

Let A, B, and C be matrices and a, b, and c be scalars. Assume that the size of the matrices are such that the operations can be performed.

Properties of Matrix Addition and scalar Multiplication

1. $A + B = B + A$ *Commutative property of addition*
2. $A + (B + C) = (A + B) + C$ *Associative property of addition*
3. $A + O = O + A = A$ (*where O is the zero matrix*)
4. $A + (-A) = 0$
5. $c(A + B) = cA + cB$ *Distributive property of addition*
6. $(a + b)C = aC + bC$ *Distributive property of addition*
7. $(ab)C = a(bC)$
8. $1C = C$

$\mathbb{R}^{m\times n}$ is a Vector Space

- $\mathbb{R}^{m\times n}$ is the set of all $m \times n$ matrices. (So, here a "vector" is one $m \times n$ matrix)
- Addition: $A + B = (a_{ij}) + (b_{ij}) = (a_{ij} + b_{ij})$ [the sum is still an $m \times n$ matrix]
- Scalar multiplication: $cA = c(a_{ij}) = (ca_{ij})$. [It is still an $m \times n$ matrix]
- The axioms and closure property hold, and therefore $\mathbb{R}^{m\times n}$, with matrix addition
and scalar multiplication, is a vector space.

1. **x** + **y** = **y** + **x** $\quad (a_{ij}) + (b_{ij}) = (a_{ij} + b_{ij})$
2. (**x** + **y**) + **z** = **x** + (**y** + **z**) $\quad \big((a_{ij}) + (b_{ij})\big) + (c_{ij}) = (a_{ij}) + \big((b_{ij}) + (c_{ij})\big)$
3. **x** + **0** = **0** + **x** = **x** $\quad (a_{ij}) + (0) = (0) + (a_{ij}) = (a_{ij})$
4. **x** + (–**x**) = **0** $\quad (a_{ij}) + (-a_{ij}) = (0)$
5. α(**x** + **y**) = α **x** + α **y** $\quad \alpha\big((a_{ij}) + (b_{ij})\big) = \alpha\,(a_{ij}) + \alpha(b_{ij})$
6. ($\alpha + \beta$) **x** = α **x** + β **x** $\quad (\alpha + \beta)(a_{ij}) = \alpha\,(a_{ij}) + \beta(b_{ij})$
7. $\alpha\beta$(**x**) = ($\alpha\beta$)**x** $\quad (\alpha\beta)(a_{ij}) = \alpha(\beta(a_{ij}))$
8. 1 **x** = **x** $\quad 1\,(a_{ij}) = (a_{ij})$

$C[a, b]$ is a Vector Space

- $C[a, b]$ is the set of all real-valued continuous functions defined on the interval $[a, b]$. Vectors in this case are continuous real-valued functions.
- For $f, g \in C[a, b]$ and scalar α, define
  - addition: $(f + g)(x) = f(x) + g(x)$
  - scalar multiplication: $(\alpha f)(x) = \alpha f(x)$.
- Check that axioms and closure property hold.

C1. $\alpha \mathbf{x} \in V$.
C2. $\mathbf{x} + \mathbf{y} \in V$.
A1. $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$
A2. $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$
A3. $\mathbf{x} + \mathbf{0} = \mathbf{0} + \mathbf{x} = \mathbf{x}$
A4. $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$

A5. $\alpha(\mathbf{x} + \mathbf{y}) = \alpha\,\mathbf{x} + \alpha\,\mathbf{y}$
A6. $(\alpha + \beta)\,\mathbf{x} = \alpha\,\mathbf{x} + \beta\,\mathbf{x}$
A7. $\alpha\beta(\mathbf{x}) = (\alpha\beta)\mathbf{x}$
A8. $1\,\mathbf{x} = \mathbf{x}$

P_n is a Vector Space

- **Definition:** a polynomial of degree k $(k \geq 0)$ is a function of the form

$p(x) = a_0 + a_1x + a_2x^2 + \ldots + a_kx^k$ where $a_k \neq 0$.

0(x) = 0 is also considered to be a polynomial.

- Example:
  - Polynomials of degree 0 (non-zero constants) 1, 2, -3
  - Polynomials of degree 1 (linear functions): x, 2x-1, 3x+8,-100x
  - Polynomials of degree 2 (quadratic functions): $x^2, 2x^2, 3x^2 + 2x + 1$
- **Definition:** P_n is the set of all polynomials of degree **strictly less than n**. Vectors in this case are polynomial functions of the form $p(x) = a_0 + a_1x + a_2x^2 + \ldots + a_{n-1}x^{n-1}$. For $p, q \in P_n$ and scalar α, define addition by $(p + q)(x) = p(x) + q(x)$ and scalar multiplication by $(\alpha p)(x) = \alpha p(x)$.

P_n is a Vector Space

- Definition: P_n is the set of all polynomials of degree **strictly less than n**. Vectors in this case are polynomial functions of the form $p(x) = a_0 + a_1x + a_2x^2 + \ldots + a_{n-1}x^{n-1}$. For $p, q \in P_n$ and scalar α, define addition by $(p + q)(x) = p(x) + q(x)$ and scalar multiplication by $(\alpha p)(x) = \alpha p(x)$.
- Check that axioms and theorem hold.

C1. $\alpha\mathbf{x} \in V$.

C2. $\mathbf{x} + \mathbf{y} \in V$.

A1. $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$

A2. $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$

A3. $\mathbf{x} + \mathbf{0} = \mathbf{0} + \mathbf{x} = \mathbf{x}$

A4. $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$

A5. $\alpha(\mathbf{x} + \mathbf{y}) = \alpha\,\mathbf{x} + \alpha\,\mathbf{y}$

A6. $(\alpha + \beta)\,\mathbf{x} = \alpha\,\mathbf{x} + \beta\,\mathbf{x}$

A7. $\alpha\beta(\mathbf{x}) = (\alpha\beta)\mathbf{x}$

A8. $1\,\mathbf{x} = \mathbf{x}$

Example: NOT a Vector Space

- The set V of all polynomials of degree **exactly** 3, with the usual addition and scalar multiplication, **IS NOT** a vector space. Which of the closure properties and axioms fails?
- Axioms that hold: A1, A2, A5, A6, A7, A8
- Axioms that fail:
  - The zero polynomial 0(x) does not have degree 3, so 0(x) is not in the set V. So A3 and A4 fails.
  - C1 fails because if α=0, then αx^3=0 which is not in V.
  - C2 fails because x^3 and $-x^3 + x^2$ are both in V, but $x^3 + \left(-x^3 + x^2\right) = x^2$ has degree 2, so it is not in V.

C1. $\alpha \mathbf{x} \in V$.
C2. $\mathbf{x} + \mathbf{y} \in V$.
A1. $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$
A2. $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$
A3. $\mathbf{x} + \mathbf{0} = \mathbf{0} + \mathbf{x} = \mathbf{x}$
A4. $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$

A5. $\alpha(\mathbf{x} + \mathbf{y}) = \alpha\,\mathbf{x} + \alpha\,\mathbf{y}$
A6. $(\alpha + \beta)\,\mathbf{x} = \alpha\,\mathbf{x} + \beta\,\mathbf{x}$
A7. $\alpha\beta(\mathbf{x}) = (\alpha\beta)\mathbf{x}$
A8. $1\,\mathbf{x} = \mathbf{x}$

Theorem

If V is a vector space and $\mathbf{x} \in V$, then

1. $0\mathbf{x} = \mathbf{0}$
2. $\mathbf{x} + \mathbf{v} = \mathbf{0}$ implies that $\mathbf{v} = -\mathbf{x}$
3. $(-1)\mathbf{x} = -\mathbf{x}$.

Proof

1. $x=(1+0)x=1x+0x=x+0x$, therefore $0x=0$
2. $x+v=0$, then $-x=-x+0=-x+(x+v)=v$
3. $0=0x=(1-1)x=1x+(-1)x$, $x+(-1)x=0$, therefore $(-1)x=-x$

Let R^+ denote the set of positive real numbers. Define the operation of scalar multiplication, denoted $\circ$, by

$$\alpha \circ x = x^{\alpha}$$

For each $x \in R^+$ and for any real number α. Define the operation of addition, denoted $\oplus$, by

$$x \oplus y = x \cdot y \text{ for all } x, y \in R^+$$

Thus for this system the scalar product of -3 times $\frac{1}{2}$ is given by

$$-3 \circ \frac{1}{2} = \left(\frac{1}{2}\right)^{-3} = 8$$

And the sum of 2 and 5 is given by

$$2 \oplus 5 = 2 \cdot 5 = 10$$

Is R^+ a vector space with these operations? Prove your answer.

Yes. Proof:

C1. Let $x, y \in \Re^+$. Then $x \cdot y$ are positive real numbers. So $x \oplus y = x \cdot y$ is a positive real number and is in $\Re^+$.

C2. Let $x \in \Re^+$ and $\alpha \in \boldsymbol{R}$. Then x is a positive real number. So $\alpha \circ x = x^\alpha$ is a positive real number and is in $\Re^+$.

A1. $x \oplus y = x \cdot y = y \cdot x = y \oplus x$

A2. $(x \oplus y) \oplus z = x \cdot y \cdot z = x \oplus (y \oplus z)$

A3. Since $x \oplus 1 = x \cdot 1 = x$ for all x, it follows that 1 is the zero vector.

A4. Let

$$-x = -1 \circ x = x^{-1} = \frac{1}{x}$$

It follows that

$$x \oplus (-x) = x \cdot \frac{1}{x} = 1 \quad \text{(the zero vector)}.$$

Therefore $\frac{1}{x}$ is the additive inverse of x for the operation $\oplus$.

A5. $\alpha \circ (x \oplus y) = (x \oplus y)^\alpha = (x \cdot y)^\alpha = x^\alpha \cdot y^\alpha$

$\alpha \circ x \oplus \alpha \circ y = x^\alpha \oplus y^\alpha = x^\alpha \cdot y^\alpha$

A6. $(\alpha + \beta) \circ x = x^{(\alpha+\beta)} = x^\alpha \cdot x^\beta$

$\alpha \circ x \oplus \beta \circ x = x^\alpha \oplus x^\beta = x^\alpha \cdot x^\beta$

A7. $(\alpha\beta) \circ x = x^{\alpha\beta}$

$\alpha \circ (\beta \circ x) = \alpha \circ x^\beta = (x^\beta)^\alpha = x^{\alpha\beta}$

A8. $1 \circ x = x^1 = x$

Since all eight axioms hold, R^+ is a vector space under the operations of $\circ$ and $\oplus$.

Summary

- Vector spaces are defined to be sets with addition and scalar multiplication, satisfying closure and other 8 axioms.
- To see if a set forms a vector space under addition and scalar multiplication, check 2 closure properties and 8 axioms.
- Besides the vector space of $\mathbb{R}^n$, many useful sets are vector spaces, including
  - The set of all matrices of size $m \times n$
  - The set of all continuous functions defined on [a, b]
  - The set of all polynomials with degree $< n$.